Bank Of Tanzania

Real Time Supervision Information System

RTSIS SANDBOX GUIDE

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Chapter 1

Introduction

1.1 Overview

This document complements RTSIS API Integration Guide to describe how Financial Service Provider (FSP) can connect and use RTSIS APIs on sandbox environment. RTSIS Sandbox exposes API services that allow FSPs to submit data on assets, liabilities, equities, off balance sheet and income statement data.

1.2 Aim and Intended Audience

The aim of this document is to provide technical guidance and examples to the Financial Service Providers (FSP) on how to test RTSIS APIs on sandbox.

Chapter 2

2 RTSIS API Developer Portal

RTSIS API Developer Portal is a portal for FSPs to explore the APIs, and generate API access token and perform tests on apis. Below are the steps to study the API and create APIs access token for connecting to RTSIS API Gateway and submitting the requests.

2.1 Login And Access Token

2.1.1 Login

1. Login to API Manager Developer portal using username and password provided
URL: https://suptech-wso2-dev.bot.go.tz:9445/devportal/
Username: (To be shared separately)
Password: (To be shared separately)

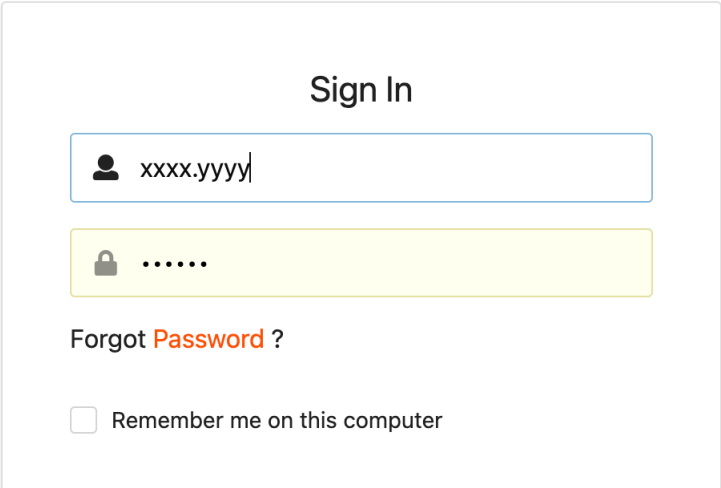

Figure 2.1: Login Page

2.1.2 Access Token

- Click Applications Tab

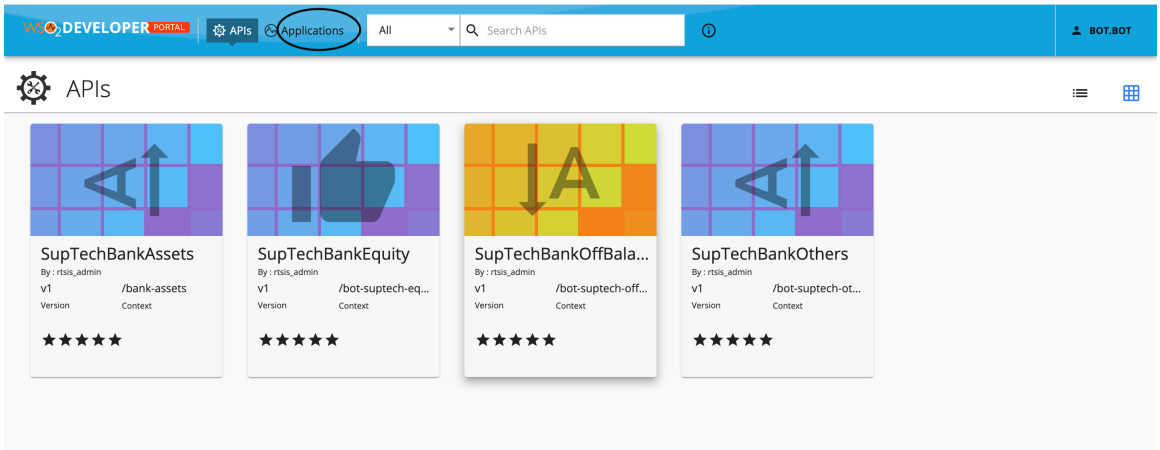

Figure 2.2: Applications

- Click Add New Application

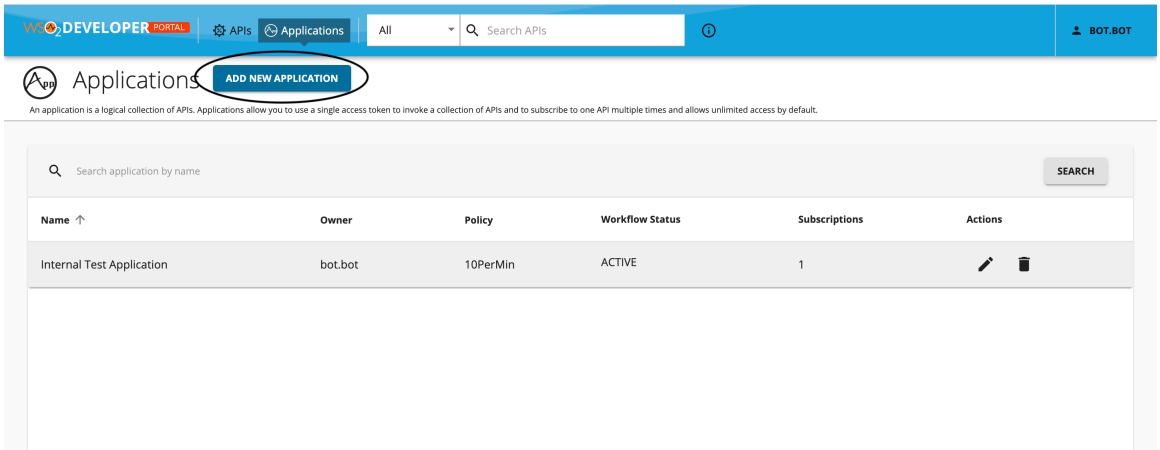

Figure 2.3: Add new applications

- Enter application detail and click 'Save' (Application name and Application Description). Set application quota to 'Unlimited' ** Application name can be any name that represent your application. The name has no effect on API integration

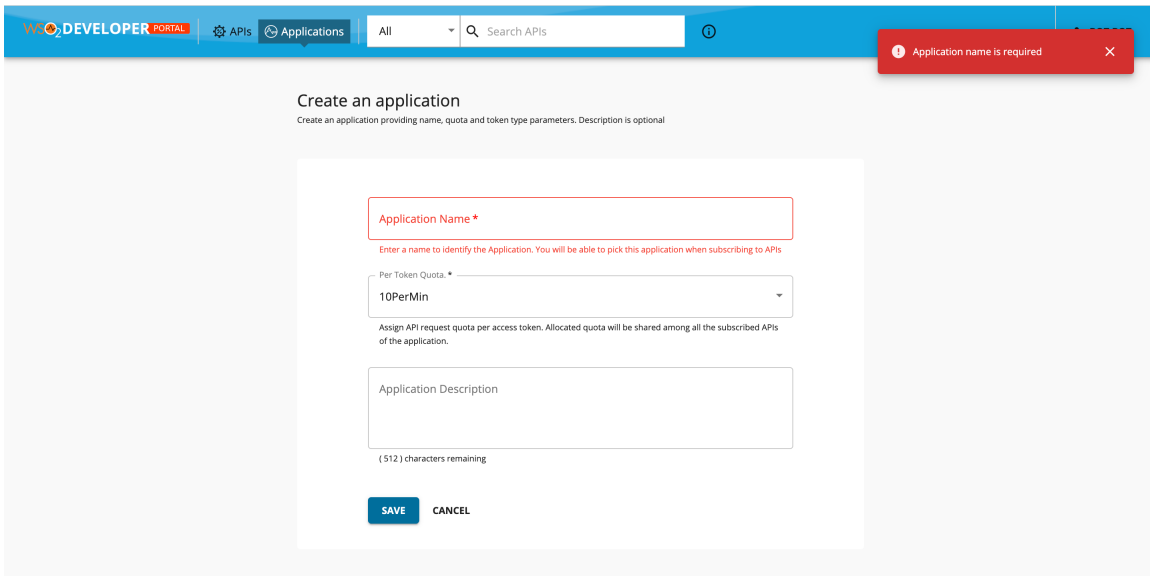

Figure 2.4: Setting Application

- Click on 'Subscriptions' tab then 'Subscribe APIs' to subscribe your created application to published APIs

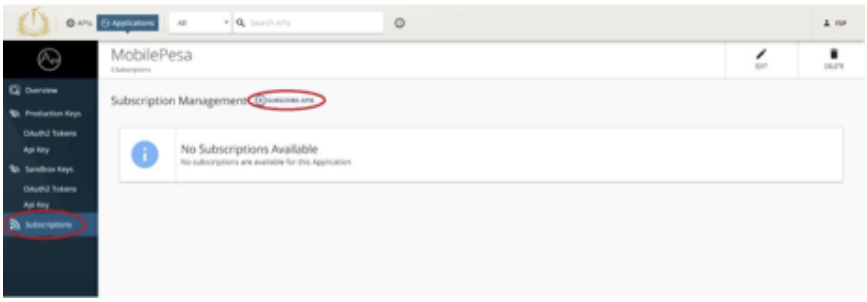

Figure 2.5: Subscriptions

- Click 'Subscribe' to subscribe to the listed APIs

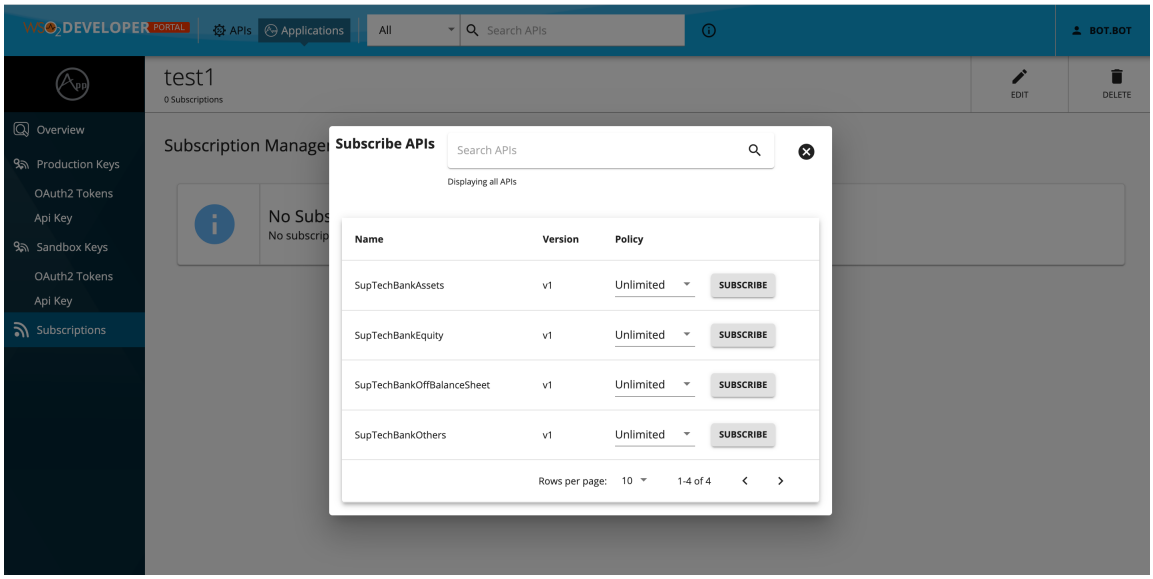

Figure 2.6: Subscribe

- Create key and secret for this application by clicking on 'Production Keys' and then 'Generate Keys'

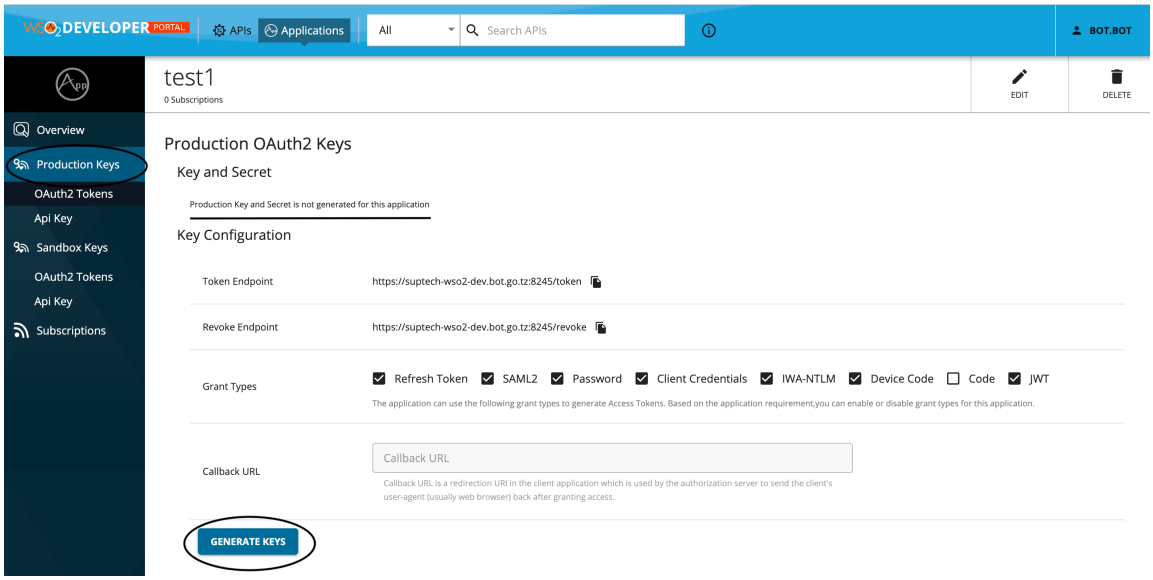

Figure 2.7: Generate Token

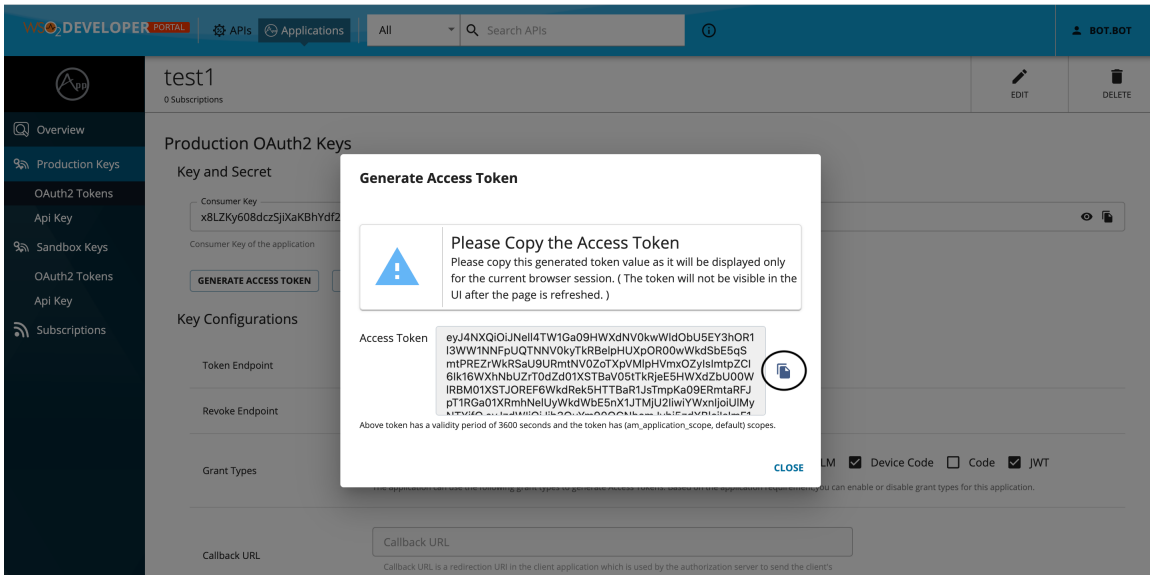

Figure 2.8: Use Token

- Click on Apis To Explore apis and Click Try Out to try apis

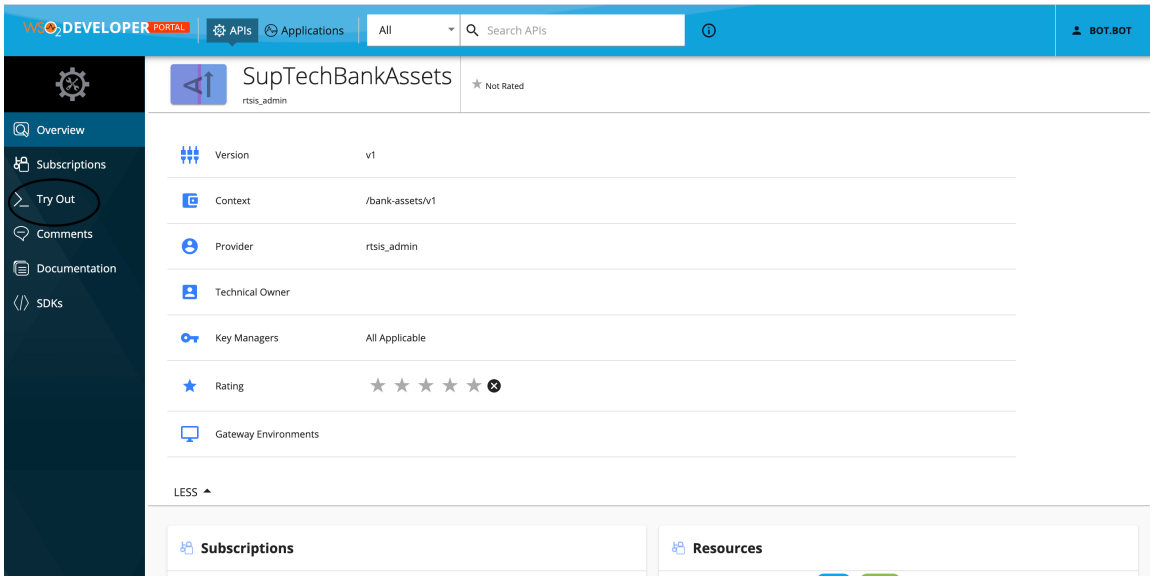

Figure 2.9: Use Token

- Select previously created application (Step 4) and Click on 'Get test key' for testing

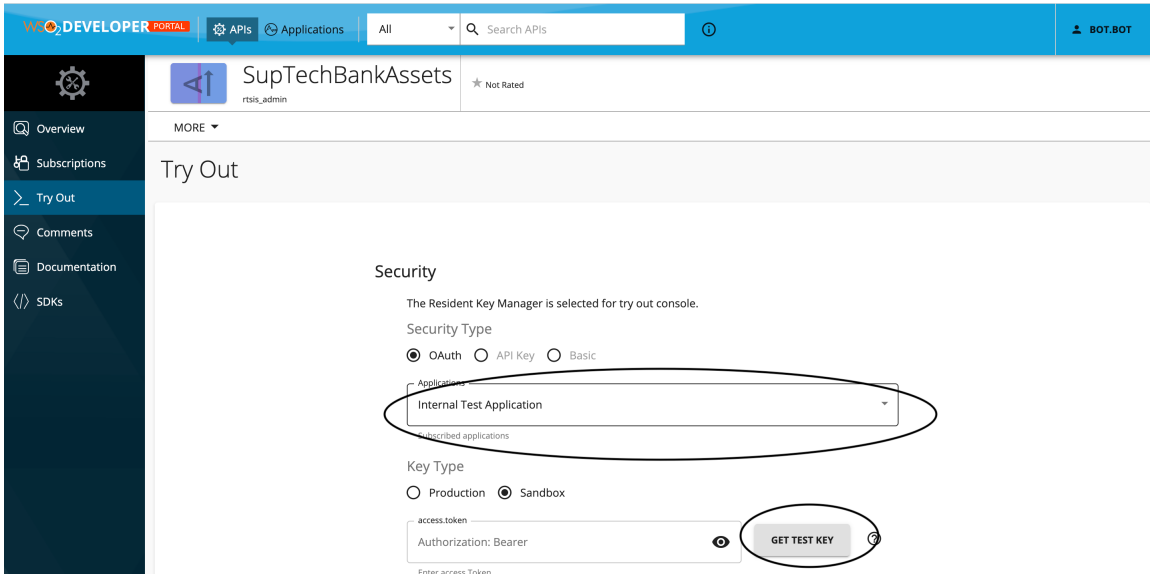

Figure 2.10: Generate Token For Test

- Scroll down to the listed APIs and select one for testing (for this demonstrations asset-loan API will be used). You may generate API Swagger to study the API by clicking on 'Swagger' download link

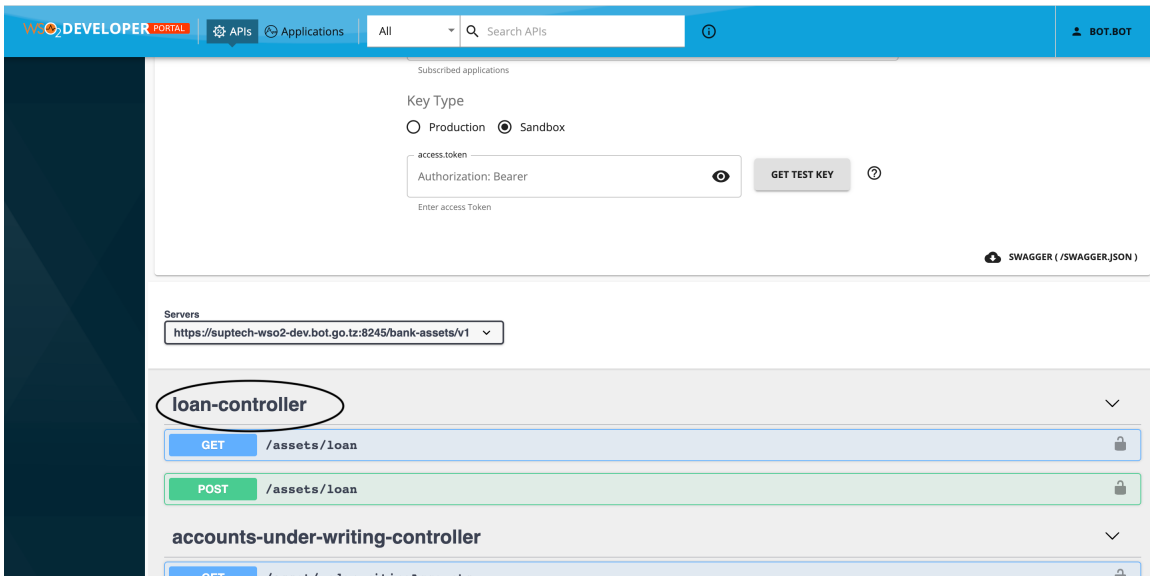

Figure 2.11: Loan Get Try out

- client Try Out And fill all required headers

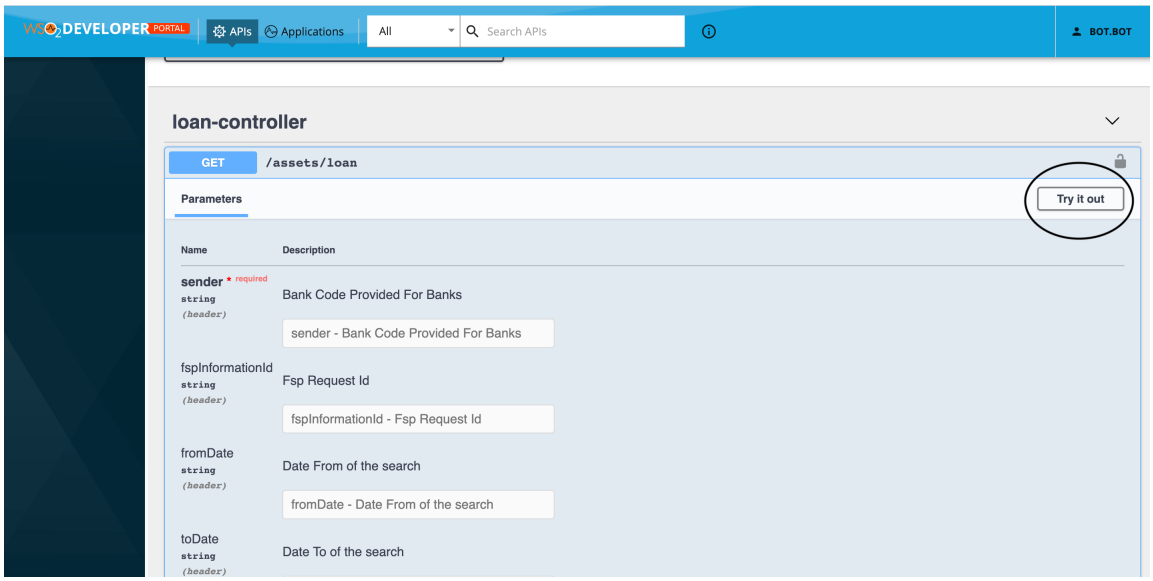

Figure 2.12: Loan Get Try out Execute

- Fill headers information and click execute

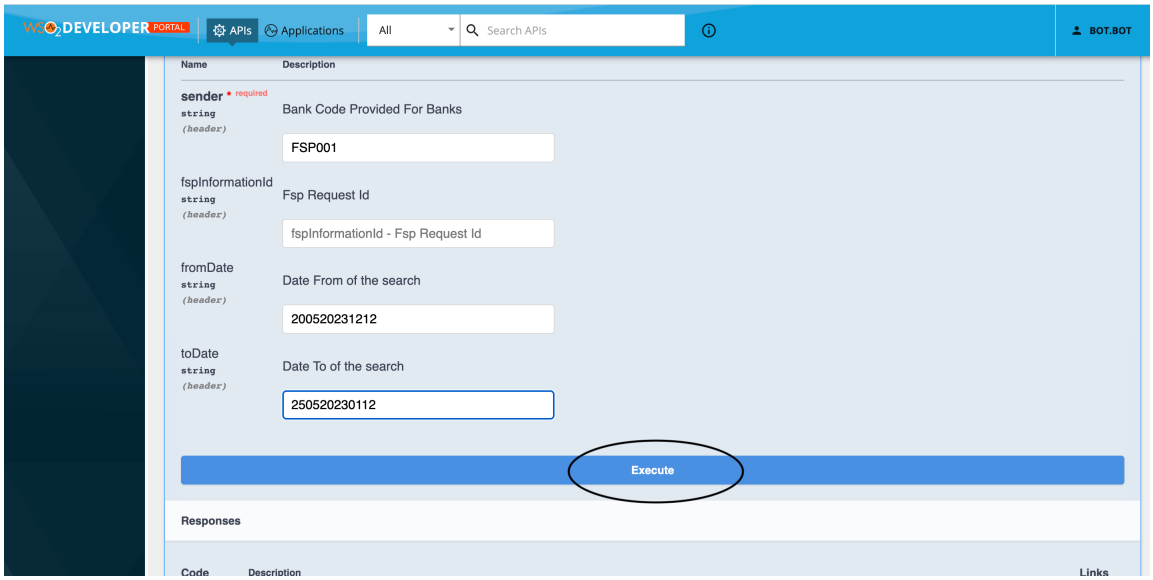

Figure 2.13: Api Request

- API Response is displayed together with Curl and Request URL

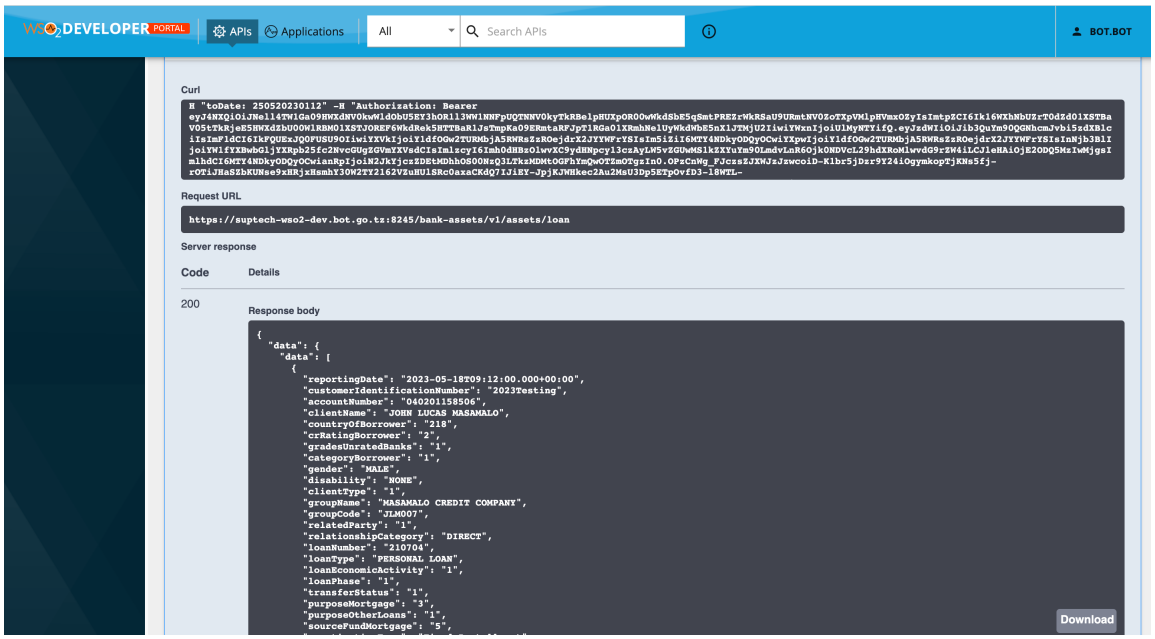

Figure 2.14: Api Response

Chapter 3

Participant Fsp Parameters

3.1 Sender Code/Fsp Code

One of the parameters required for submission of GET and POST requests is the sender code which represents Fsp Code which is shared on the table on the next page which provides code for Fsp. The table is prone to modification upon the addition and removal of Fsps.

Table 3.1: FSP Names and Codes

FSP Name	FSP Code
ABSA BANK	020
ACCESS BANK (T) LTD	035
AKIBA COMMERCIAL BANK	012
AMANA BANK	046
AZANIA BANCORP BANK	031
BANC ABC	034
BANK OF AFRICA (TANZANIA) LTD	009
BANK OF BARODA (T) LTD.	029
BANK OF INDIA (T) LTD	036
CANARA BANK LTD	058
CHINA DASHENG BANK LTD	065
CITIBANK TANZANIA LIMITED	008
CRDB LIMITED	003
DCB COMMERCIAL BANK	024
DIAMOND TRUST BANK	011
ECOBANK TANZANIA LTD	040
EQUITY BANK (T) LTD	047
EXIM BANK	013
FINCA MICROFINANCE BANK	056
FIRST NATIONAL BANK (T) LTD	045
GUARANTY TRUST BANK TANZANIA LTD	061
HABIB AFRICAN BANK	018
I AND M BANK (T) LTD	021
INTERNATIONAL COMMERCIAL BANK (T)	019
KCB BANK TANZANIA LIMITED	017
KILIMANJARO CO-OPERATIVE BANK	014
LETSHEGO BANK TANZANIA LTD CLEARING	044
MAENDELEO BANK	051
MKOMBOZI COMMERCIAL BANK PUBLIC LTD	039
MUCOBA BANK PLC	064
MWALIMU COMMERCIAL BANK OF TANZANIA	059
MWANGA HAKIKA MICROFINANCE BANK	042
NATIONAL MICROFINANCE BANK	016
NBC LIMITED	015
NCBA BANK TANZANIA LTD	023
PEOPLES' BANK OF ZANZIBAR	004
STANBIC BANK TANZANIA	006
STANDARD CHARTERED BANK	005
TANZANIA AGRICULTURAL DEVELOPMENT	052
TANZANIA COMMERCIAL BANK PLC	048
UCHUMI COMMERCIAL BANK	032
UNITED BANK FOR AFRICA	038
YETU MICROFINANCE BANK PLC	063
BANK OF TANZANIA	001
MIC TANZANIA PLC	501
ZANZIBAR TELECOM PLC	502
VODACOM TANZANIA PLC	503
AIRTEL TANZANIA PLC	504
TANZANIA TELECOMMUNICATIONS CORPORATION	505
VIETTEL TANZANIA PLC	506

Chapter 4

Digital Signature

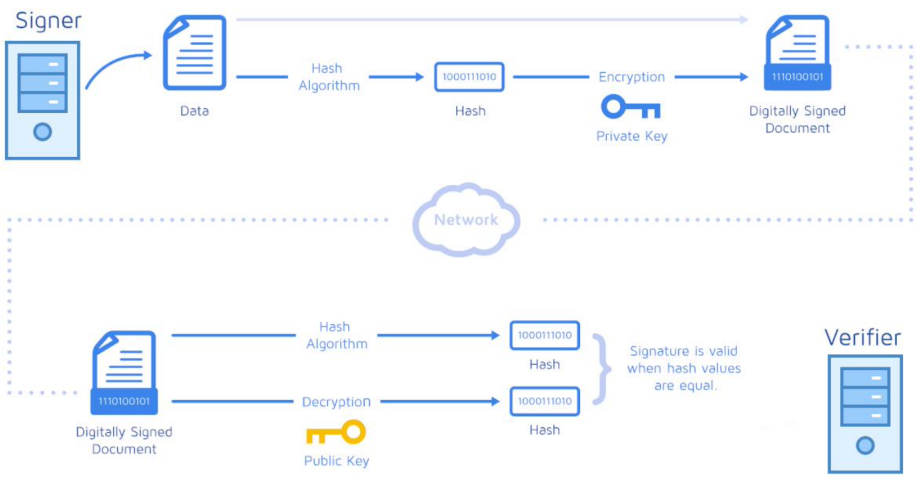

Figure 4.1: Digital Signature

4.1 Overview

Digital signature will be used for messages exchange between the FSP and BOT. The FSP will be required to apply for digital signature from Certificate Authority (CA). Digital signatures will also be used for message(s) sharing. Message(s) must previously be signed by the Participant's private key and verified using participant's Public Key.

4.2 Digital Signature

The security architecture used by the RTSIS is based on Public Key Infrastructure (PKI) technology in which two keys (Public and Private) are used to provide the highest levels of security and universal acceptance. There are typically three algorithms involved with the digital signature process:

- Key generation — This algorithm provides a private key along with its corresponding public key from CA.
- Signing — This algorithm produces a signature upon receiving a private key and the message that is being signed.
- Verification — This algorithm checks for the authenticity of the message by verifying it along with the signature and the public key.

4.3 Digital Signature File Format

RTSIS accepts and shares public and private keys in Public-Key Cryptography Standards version 12 (PKCS #12). The expected filename extension for PKCS #12 files is .p12 or .pfx. For example, fspABCPublicKey.pfx or fspABCPublicKey.p12

4.4 Signature Algorithm

The supported signature algorithms are

1. SHA256withRSA
2. SHA256withECDSA

4.5 Signed Final Payload Format

RTSIS expects message to be digitally signed and obey the following rules:

- sample content before signing example from balanceBot post api

```
"balanceBOT" : [ {
    "accountNumber" : "992468140",
    "accountName" : "CRDB BANK TZ LTD",
    "usdAmount" : 1000.59,
    "tzsAmount" : 1000.59,
    "orgAmount" : 1000.59,
    "reportingDate" : "101020221000",
    "maturityDate" : "101020221000",
    "transactionDate" : "101020221000",
    "currency" : "1",
    "subAccountType" : "Other Accounts",
    "accountType" : "1"
  } ],
```

- Final payload after signing

```
    {
  "balanceBOT" : [ {
    "accountNumber" : "992468140",
    "accountName" : "CRDB BANK TZ LTD",
    "usdAmount" : 1000.59,
    "tzsAmount" : 1000.59,
    "orgAmount" : 1000.59,
    "reportingDate" : "101020221000",
    "maturityDate" : "101020221000",
    "transactionDate" : "101020221000",
    "currency" : "1",
    "subAccountType" : "Other Accounts",
    "accountType" : "1"
  } ],
  "signature" : "S2bqTHPlpwpcl/OwjzLKB
  Buwa6VVWtC9hM07yfrKK4DF7q909UUVidR/
qooCfaPSfy61AL2YSZ424ouizFsKMA2WGx9nz
```

```
JfYRNQzgwf8zBSOphl5cNwSjoJAvBejXlgIc
OLjKzPcfSsXqXraEKoptRpDId5kFqRTYgp4u3
C4CqdgJRaS2oGvtz6/yNshPvExM3TQ00LdN9
QuDU+kPgmDCFy1QcyQq2RsIY3S2KvMxWAq/
iWd83ChcdTabghFpOnlCSg3awxZTuqZvvxYpCl
LSRUfbTGsN31V8KgP5rwQJRMi4cBERwgbRozAln
pPbgUhhsj4Ux9fUI+LClsIhXDYUA=="
}
```


NOTE: In order to enable easy testability direct from the developer portal only a few endpoints will be enabled with signature verification in the early stages. But it's important to note that later all Apis will be enabled with signature verification there fore implementing signature in the early stages is very important. sampling codes for signing for all languages will be provided in our development communications groups.

Chapter 5

More Examples

5.1 Post Balance Bot

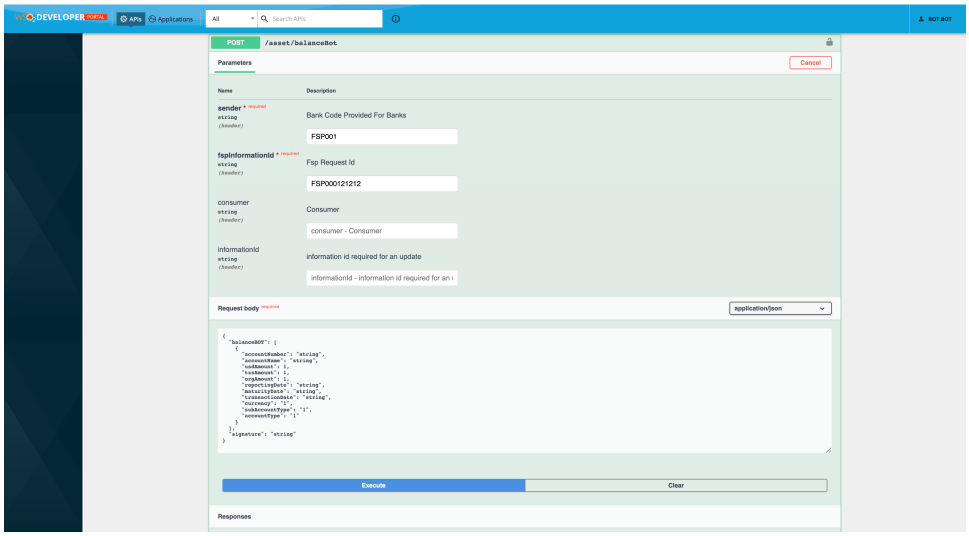

Figure 5.1: Post Balance Bot Request

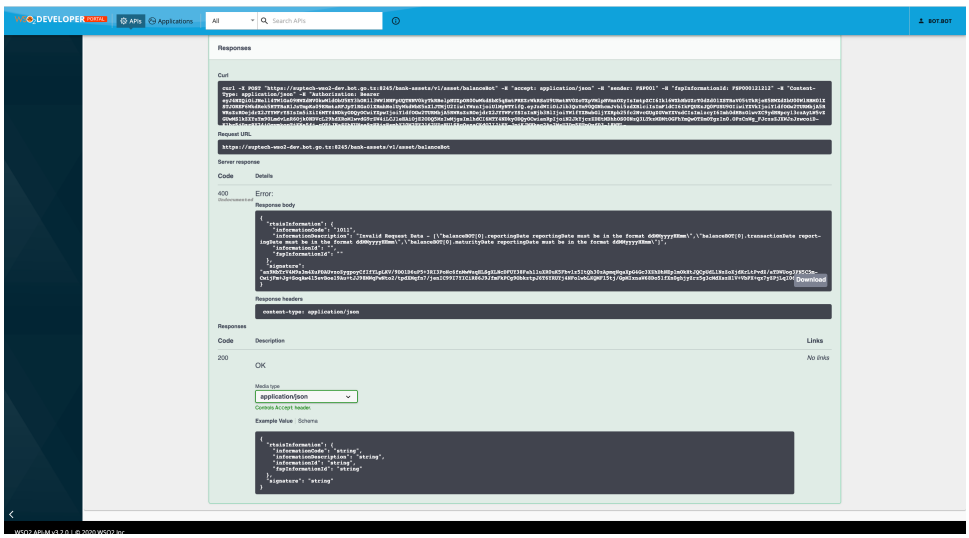

Figure 5.2: Post Balance Bot Response